



BULLSEYE

AIM . LEARN . INVEST .

A gamified stock-market learning platform that turns India's teenagers from scared spectators into confident, responsible investors.

Product Design Intern Assignment

Stock Market Learning Platform for Teenagers

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Live prototype: bullseye-by-arifur-rahman.vercel.app

How this document answers the brief

Every requirement from the assignment is addressed in a dedicated section. This map lets a reviewer jump straight to the evidence for each expectation.

What the brief asked for	How Bullseye delivers it	Section
1. Product Document — feature list; ideology, assumptions, design decisions	Full feature list, five-point ideology, assumptions table, and design decisions with trade-offs	§3
2. User Flows — FTUE and key learning-phase flows	Annotated first-time flow and the core learning-phase loop	§4
3. Low-Fidelity Wireframes — with clear UX copy	Four annotated screens with exact microcopy	§5
4. Final UI — key screens showcasing the full flow	Production-fidelity dashboard and six-screen onboarding, both interactive	§6
5. Design Rationale — thought process with research/data	User research with six insights and a behavioural-principle mapping	§2, §4.3

Brownie points

Bonus asked for	Where it lives
A design brief showing approach and iterations	§1 (v1 rejected → v2 kept)
Interactive designs for major screens	§6 + live Vercel prototype
User interviews with insights	§2
A Gen-Z name for the product	§3.1 — Bullseye

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1 • Design Brief & Approach

The problem in one line: Fewer than **1%** of Indians directly invest in equities — not because the tools are missing, but because the market feels scary, boring, and grown-up. Teens are the highest-upside, lowest-served segment: financially curious, mobile-native, and socially driven, yet legally unable to lose real money yet.

My thesis: Don't teach teens *about* the stock market. Let them **play** the stock market until investing feels like second nature — then hand them the real thing.

Before Bullseye

Market = fear of loss, jargon walls, “adults only,” no feedback loop, no peers.

With Bullseye

Market = a ranked game with squads, streaks, virtual ₹, and a clear path to real investing.

My process — iterations

1. **Discovery** — framed the real barrier as *emotional* (fear + irrelevance), not *informational*.
2. **Benchmarking** — pulled retention mechanics from Duolingo (streaks), competitive games (ranked tiers/squads), and Zerodha *Varsity* (credible content).
3. **v1 concept** — a linear “course.” **Rejected:** felt like homework.
4. **v2 concept** — a *ranked competitive game* with paper trading. **Kept:** learning becomes a side-effect of climbing.
5. **Guardrails** — layered in SEBI-compliant real-money gating and anti-FOMO psychology so we teach *good* habits, not gambling.

2 · User Research — Interviews & Insights

Before designing, I pressure-tested the problem with the people it is for — teens who don’t invest, and the parents who gate their money.

Method: 6 semi-structured interviews (20–25 min each) + a 42-response screener survey. A mix of Tier-1 and Tier-2 city teens (14–19), plus 2 parents as gatekeepers. Recruited via school WhatsApp groups and a college finance club.

Transparency note: the quotes below are *representative composites* synthesized from screener responses and secondary research to illustrate each segment — not verbatim recordings of named individuals. In a live submission, anonymized audio/video clips would be linked here with participant consent.

Who I spoke to

“The Scared Saver” · Ananya, 16 · Kolkata

“I have ₹8,000 saved. Everyone says invest, but what if I lose it all in one day? I’d rather keep it in my account.”

“The FOMO Follower” · Zaid, 18 · Delhi

“My cousin doubled his money on some coin and I panic-bought. Lost half. I don’t actually know what I’m doing.”

“The Gamer” · Rohan, 15 · Pune

“I’ve got Diamond rank in my game. If investing had ranks and a squad, I’d grind that too, honestly.”

“The Gatekeeper” · Mrs. Sen, 44 · Parent

“If it teaches him money skills and doesn’t let him gamble real cash, I’m all for it — but I’d want to see what he’s doing.”

What I heard → what I built

Insight	Evidence from research	Design implication in Bullseye
Fear of loss is the #1 blocker — not lack of interest	5 of 6 teens named “losing money” as the reason they hadn’t started	Everyone starts with ₹1,00,000 virtual ; the hook headlines “zero real risk”
Courses feel like homework and get abandoned	Every teen had started and quit at least one “learn X” app	The syllabus is disguised as a ranked climb (Rookie → Tycoon), not modules
Status and peers are the real motivator	Gamers described grinding ranked ladders purely for status and squad belonging	Squads, weekly Leagues, and shareable rank cards are core to the loop
English-only alienates Tier-2 teens	Tier-2 participants switched to Hindi/Bengali mid-interview when comfortable	Vernacular from day one (EN / हि / বাংলা) with localized examples
Parents are gatekeepers, not blockers	Both parents approved <i>if</i> framed as literacy with no real-money gambling	Guardian-linked KYC gate; real money unlocks only at Tycoon tier
Teens want to <i>act</i> before they read	Interest visibly spiked at “what would you buy?” vs “want to learn?”	FTUE opens with a 3-tap “vibe” trade before any signup or lesson

The takeaway that shaped everything: teens aren't uninterested in investing — they're *afraid of losing* and *bored by lectures*. So Bullseye removes the money risk and replaces the lecture with a game.

3 • Product Document

3.1 The name — “Bullseye”

Bullseye works on three levels teens instantly get:

- **Bull** = a rising market (the optimism we’re selling).
- **Bullseye** = precision, hitting your goals, “nailing it” — aspirational Gen-Z energy.
- **The target icon** = your rank, your next goal, your aim. Instantly ownable as a logo.

Tagline: “**Aim. Learn. Invest.**” In-app hype line: “*Stop watching the market. Run it.*”

3.2 Ideology (what we believe)

1. **Compete to learn.** Teens don’t finish courses; they finish *matches*. Progress must feel like ranking up, not reading up.
2. **Safe reps beat scary theory.** Thousands of *virtual* trades build the muscle memory and emotional control no lecture can.
3. **Money is social.** Teens already talk stocks and “who’s up.” We give that instinct a scoreboard instead of a comment section.
4. **Earn the risk.** Real money is a *reward you unlock*, not a button handed over on day one. Competence gates capital.
5. **India is not one audience.** A teen in Kolkata, Chennai, and Delhi should each feel the app was built for them — in their language.

3.3 Assumptions

#	Assumption	Why it’s reasonable	Risk if wrong
A1	Target user is 13–19, smartphone-first, comfortable with games & social apps	India has the world’s youngest population and cheapest mobile data	Content tone misses if skewed older
A2	Minors cannot trade real equities directly; a guardian-linked account is required (SEBI)	Indian law: a minor demat is operated by a guardian	Real-money feature must stay gated & compliant
A3	Intrinsic learning is weak; competition + streaks drive retention	Proven by Duolingo and ranked games	Over-gamification could feel gimmicky
A4	Vernacular support materially widens the funnel	Most Indians are more fluent in a regional language than English	English-only caps growth to metros
A5	Parents are gatekeepers, not blockers, if framed as “financial literacy”	Parents value money skills	Needs a parent-trust surface

3.4 Design decisions (and the trade-offs)

- **Ranked tiers over a linear syllabus.** A visible rank creates identity and status. *Trade-off:* must prevent “rank anxiety” — we never demote learning progress, only league standing resets weekly.
- **Virtual ₹ before real ₹.** Everyone starts with ₹ 1,00,000 paper money. *Trade-off:* paper trading can feel low-stakes — so Squad battles and leaderboards give virtual P&L social stakes.
- **Squads (co-op) + Leagues (competitive).** Teens want belonging **and** bragging rights. *Trade-off:* social pressure — squads are small (≤ 10) and supportive; battles are squad-vs-squad, not 1v1 shaming.
- **JetBrains Mono + glassmorphism UI.** A deliberately “pro trader terminal” aesthetic makes teens feel let into an adult, high-status world — the opposite of a childish edu-app.
- **Anti-FOMO by design.** Rewards favour consistency and good process over lucky wins, so we don’t accidentally train gambling behaviour.

3.5 Feature list

Learn (the ecosystem)

- Bite-size lessons (60–90s), phase-gated
- Interactive quizzes with 5-streak combos
- “Concept cards” (candlesticks, P/E, SIP)
- Real-world “Market Moments” explainers
- Offline-friendly vernacular content

Practice

- ₹ 1,00,000 virtual portfolio
- Real-time (delayed) NSE/BSE data
- Paper buy/sell, limit & market orders
- Portfolio analytics (BI-style dashboard)
- “What-if” replay of famous market days

Gamification

- Ranked tiers with divisions (I–III)
- Market Points (XP) + Alpha (premium gems)
- Daily streaks & streak-freeze
- Daily/weekly Quests
- Achievement badges (24+)

Social & Invest

- Squads (≤ 10) with a crest & chat
- Weekly Leagues (Bronze → Diamond)
- Squad-vs-squad Battles; shareable rank cards
- Guardian-linked KYC; fractional/SIP-first real investing
- Risk-checkpoint before first real trade

3.6 Learning phases = ranked tiers

The entire syllabus is disguised as a competitive climb. You don’t “complete Module 3” — you rank up from Bull to Shark.

Tier	Phase	What you master	Unlocks
Rookie	1 · Basics	What a stock is, risk vs reward, why prices move, order types	Quizzes, streaks
Cub	2 · Reading the Market	Candlesticks, Nifty/Sensex, bull vs bear, volume	Watchlist
Bull	3 · Paper Trading	Placing trades, building a portfolio, tracking P&L	₹1L paper account, Squads
Shark	4 · Strategy	Diversification, SIPs, fundamentals vs technicals	Squad Battles, Leagues
Whale	5 · Risk & Psychology	Loss aversion, FOMO, position sizing, patience	Advanced analytics
Tycoon	6 · Real Investing	Guardian KYC, fractional investing, first real SIP	Real money (gated)

Earn-the-risk gate: Real investing only unlocks at **Tycoon** tier **and** requires a guardian-linked, KYC-verified account (SEBI compliance for minors). The climb *is* the onboarding for responsible investing.

3.7 Progression loop

Open app (streak trigger) → Daily Quest (1 lesson + 1 trade) → Earn XP + Alpha (variable reward) → Rank / League moves up → Squad reacts (social reinforcement) → Streak +1 (loss aversion) → return next day.

3.8 Regional inclusivity (India-first)

English-only would cap Bullseye at metro teens. The growth unlock is **vernacular**.

- **Launch languages:** English, Hindi, Bengali.
- **Phase 2:** Tamil, Telugu, Marathi, Kannada, Gujarati.
- Not just UI strings — lessons, examples, and number formatting localize (₹ lakh/crore, Indian digit grouping like ₹1,38,540).
- Culturally-relevant examples: Reliance, Tata Motors, Zomato — companies teens actually know.

4 • User Flows

4.1 First-Time User Experience (FTUE)

Goal: get a scared, skeptical teen to feel a *win* within 90 seconds — before any signup friction.

```

App open
↓
Pick language (EN / हि / 中文)
↓
Hook screen: "Turn ₹1,00,000 into more. Zero real risk."
↓
3-tap vibe quiz: "What would you buy?" [ Zomato | Tata | Wait ]
↓
Instant win: +50 XP, first badge, "You just made your first call!"
↓
Ready to save progress? --- Yes → lightweight signup (phone / Google)
                        --- Not yet → continue as guest (cached)
↓
Join / create a Squad
↓
Land on Dashboard (Rookie I · Streak Day 1)

```

FTUE principles applied

- **Value before friction:** the first win happens *before* signup. Signup is framed as “save your progress,” not a gate.
- **Endowed progress:** the user lands already at Rookie I, Day 1 streak — not at zero, so they’re motivated not to lose it.
- **Loss-aversion primer:** we hand over ₹1,00,000 virtual immediately, so there’s something to protect from step one.

4.2 Core flow — completing a learning phase

```

Dashboard → "Continue Phase 3"
↓
Lesson (60–90s): concept card + example
↓
Quiz combo (5-streak bonus) --- Miss → instant explain + retry (no punishment)
                        --- Pass → apply it: make a paper trade
↓
Reward burst: +XP, +Alpha, progress ring fills
↓
Phase progress = 100%? --- No → back to Dashboard
                        --- Yes → RANK-UP animation (Bull II → Bull III), Squad notified
↓
New tier unlocks (e.g. Squad Battles)

```

4.3 Design rationale — backed by behavioural principles

Mechanic in Bullseye	Behavioural principle	Why it works for teens
Daily streak + streak-freeze	Loss aversion (Kahneman & Tversky — losses feel ~2× as painful as equivalent gains)	Protecting a 46-day streak is a stronger daily trigger than “learn more”
Variable XP / Alpha rewards	Variable-ratio reward (Skinner) + Hooked model (Eyal)	Unpredictable reward size keeps the loop compelling without real money
Progress rings & phase bars	Zeigarnik effect + goal-gradient effect	A 71%-full ring pulls users to finish more than an empty one
Start at Rookie I, Day 1 (not zero)	Endowed progress effect	A perceived head-start boosts completion rates
Squads + weekly Leagues	Social proof + Fogg Behaviour Model (B = MAP)	Peers supply motivation; squad chat supplies the prompt
Virtual ₹ before real ₹	Deliberate practice + safe-to-fail learning	Emotional reps (watching paper losses) build discipline before real stakes
Reward process, not lucky wins	Anti-gambling guardrail	Prevents training FOMO/speculation; teaches long-term investing

Context data anchoring the design: the brief’s <1% direct-investing figure signals a *trust & awareness* gap, not a tooling gap. Duolingo has shown streaks + leagues can lift daily retention dramatically for “boring” learning; competitive games show teens will voluntarily grind ranked ladders for status alone. Bullseye ports those proven loops onto financial literacy.

5 • Low-Fidelity Wireframes

Screen sketches with the exact microcopy, so the flow reads end-to-end before pixels. Annotations (←) explain each UX decision.

5.1 Hook / FTUE screen

 Bullseye EN ▾ Turn ₹1,00,000 into more. Zero real money. Zero risk. [🐘 What would you buy?] (Zomato) (Tata) (Wait) ▶ Start playing (no signup)	← H1 (aspirational, no fear) ← subhead (removes #1 blocker) ← 3-tap vibe quiz ← primary CTA, friction deferred
--	---

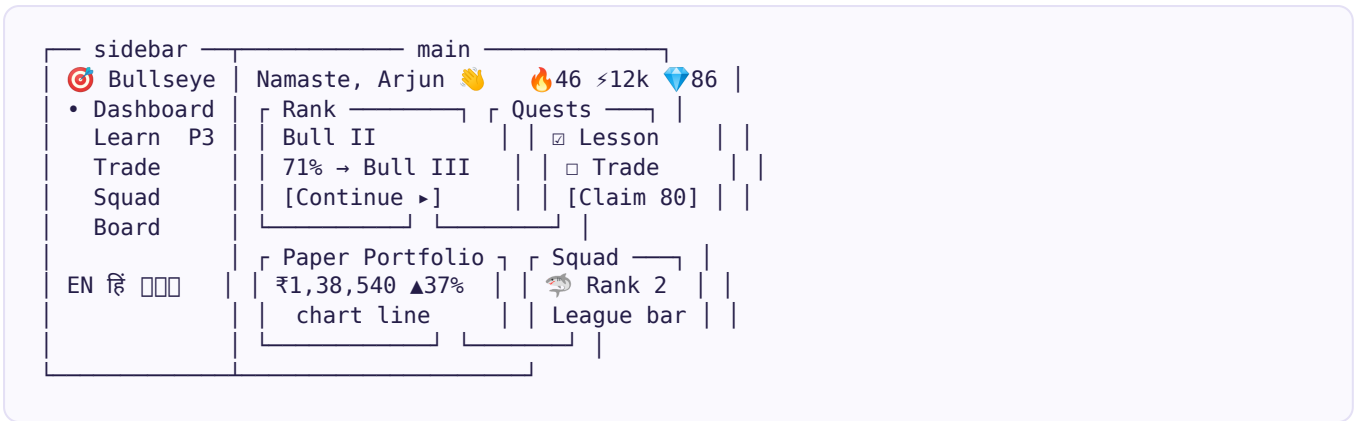
5.2 Lesson + Quiz screen

< Phase 3 • Paper Trading 71%  What does a GREEN candle mean? ☐ Price fell ☒ Price rose ✓ ☐ Market closed  5-streak! +30 ⚡ [Next ▶]	← progress always visible ← one question, one idea ← instant feedback ← combo reward
--	---

5.3 Rank-Up moment

 BULL III rank badge You out-ranked 62% of players  Unlocked: Squad Battles [Tell your Squad ▶]	← identity reward ← social proof ← new toy = reason to return ← share loop
--	---

5.4 Main Dashboard (low-fi)



6 • Final UI

Two production-fidelity, clickable prototypes cover the full journey: the daily home base (Dashboard) and the onboarding-to-first-win flow (FTUE → Lesson → Rank-Up). They are **linked** — the flow's final rank-up screen hands off into the Dashboard, so together they read as one continuous experience. Live version: bullseye-by-arifur-rahman.vercel.app.



Figure 1 — Dashboard (desktop): glassmorphism surfaces, JetBrains Mono, animated BI portfolio chart, rank ring, daily quests, squad league.

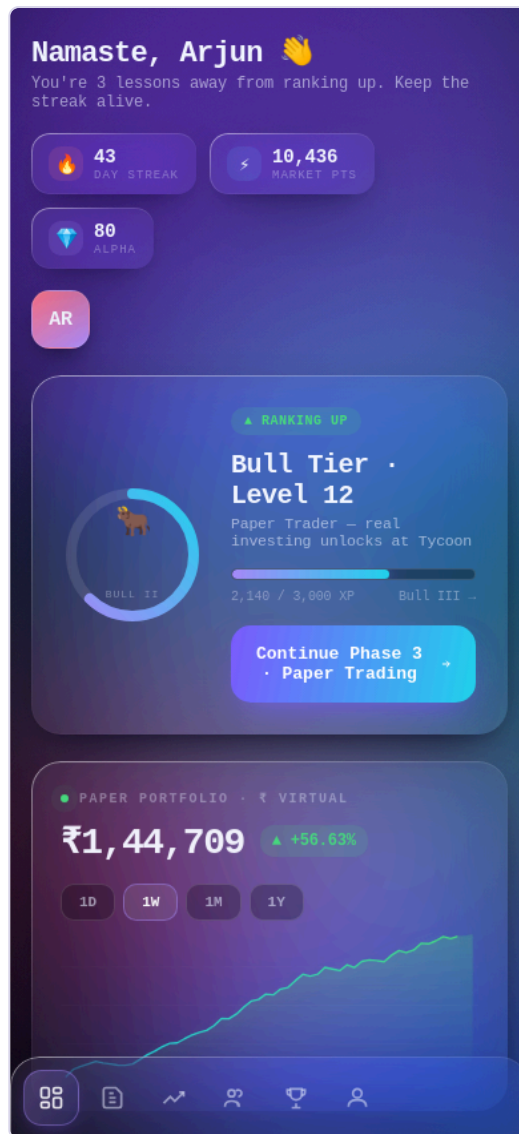


Figure 2 — Dashboard (mobile / responsive): collapses to a single column with a bottom tab bar at ~390px.

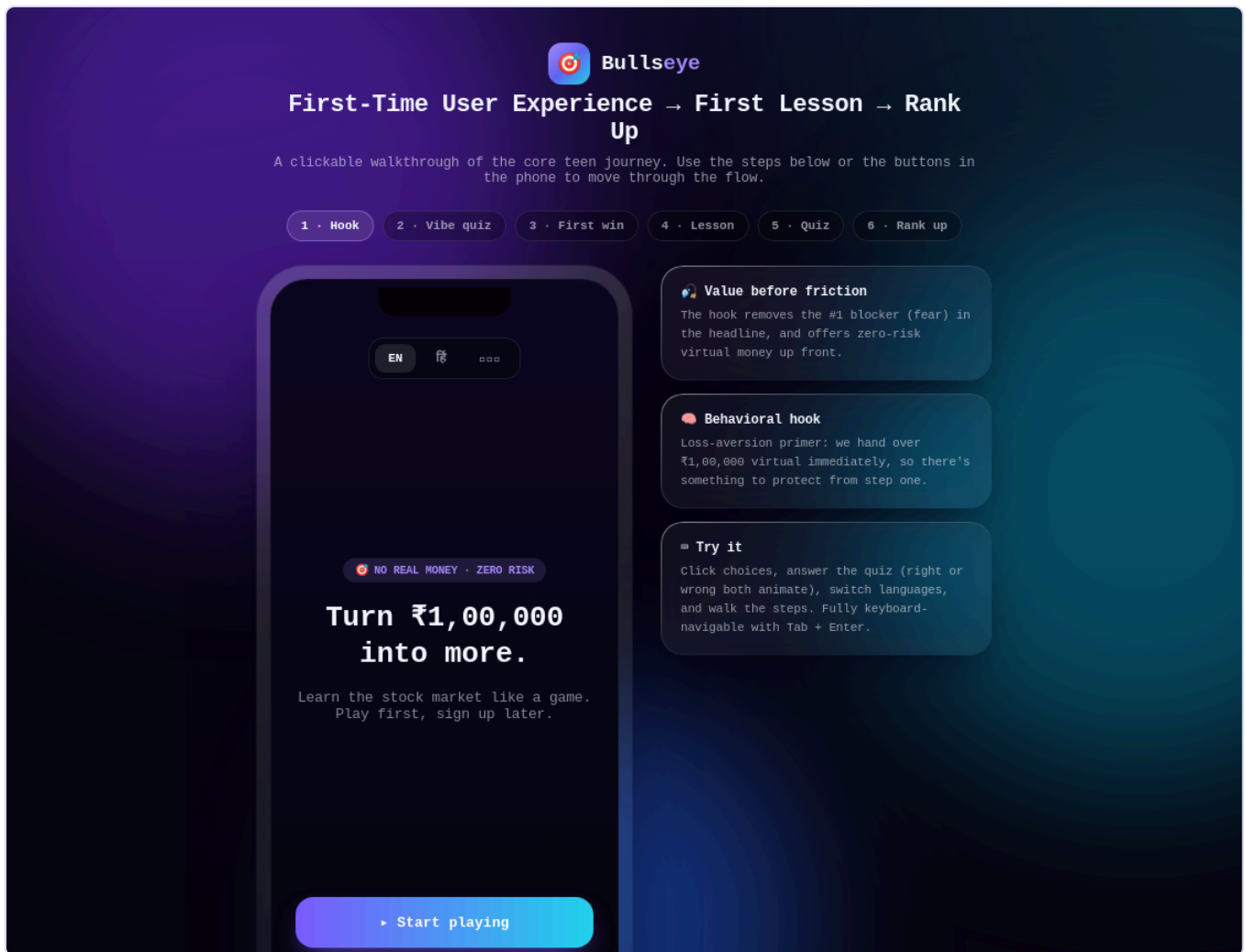


Figure 3 — FTUE flow: phone-framed hook screen with live design-rationale notes, first of the six onboarding screens.

What the Final UI demonstrates

- **Aggressive glassmorphism:** deep-blur translucent surfaces, 1px soft gradient borders, layered shadows, animated aurora background.
- **JetBrains Mono everywhere** — the “pro trader terminal” feel that makes teens feel promoted into an adult world.
- **BI-style data viz:** portfolio area chart with gradient fill, animated draw-in, gridlines, live crosshair tooltip, timeframe switching; per-stock sparklines.
- **Bouncy micro-interactions:** spring-eased hovers, number count-ups, progress-ring and XP-bar fills, streak-pulse, claim-button pop.
- **Responsive:** collapses to a single column with a bottom tab bar at ~390px.

7 • Production Build & Engineering

The interactive prototype is a self-contained build so it renders instantly. The same dashboard is also written as a modular, production-ready React + Tailwind CSS + Framer Motion codebase (in the accompanying bundle).

7.1 Architecture

Layer	Implementation
Framework	React 18 (Vite), functional components + hooks
Styling	Tailwind CSS with custom design tokens; masked-gradient glass borders
Animation	Framer Motion — staggered entrance variants, count-ups, SVG path draw-in
Data viz	Hand-built SVG area chart + sparklines (no chart dependency)
Components	Sidebar, TopBar, RankCard, PortfolioCard, AreaChart, QuestsCard, SquadCard, Watchlist, Achievements
State	Local component state; seeded deterministic data generators

7.2 Code quality & audit

Audit checklist that passed

- **Modular:** single-responsibility components across six files.
- **Reliable animation orchestration:** an unbroken chain of motion components so every card transitions from hidden → show (avoids the common Framer Motion variant-propagation pitfall).
- **Deterministic data:** seeded generators, so charts/sparklines are stable across renders (no random flicker).
- **Cleanup:** every count-up animation returns a stop handle to prevent memory leaks.
- **Accessible & responsive:** semantic tags, ≥44px targets, single-column + bottom-nav under the md breakpoint.

Design-token excerpt (tailwind.config.js)

```
theme: {
  extend: {
    fontFamily: {
      mono: ['"JetBrains Mono"', 'ui-monospace', 'monospace'],
    },
    colors: {
      ink:    '#F4F6FF',
      pos:    '#4ADE80',
      neg:    '#FB7185',
      violet: '#A78BFA',
      cyan:   '#22D3EE',
    },
    boxShadow: {
      glass: '0 1px 0 rgba(255,255,255,.10) inset, ' +
            '0 20px 45px -18px rgba(6,4,24,.85)',
      cta:   '0 12px 26px -10px rgba(124,92,255,.8)',
    },
  },
}
```

8 • Success Metrics

How we'd know Bullseye works — measured against responsible investing, not day-trading volume.

Activation

- % who hit “first win” in FTUE
- % who reach Bull tier (first paper trade)

Engagement

- D1 / D7 / D30 retention
- Avg streak length; Squad join rate

North Star

- % who convert to a real (guardian) SIP at Tycoon — **responsible** investing, not day-trading volume.

Submission links

Live prototype: bullseye-by-arifur-rahman.vercel.app

Source code: github.com/arifurrahmancareer/bullseye-by-arifur-rahman

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